Ex.no -3.1- CREATE 3D MODEL FROM ORTHOGRAPHIC PROJECTION

AIM:

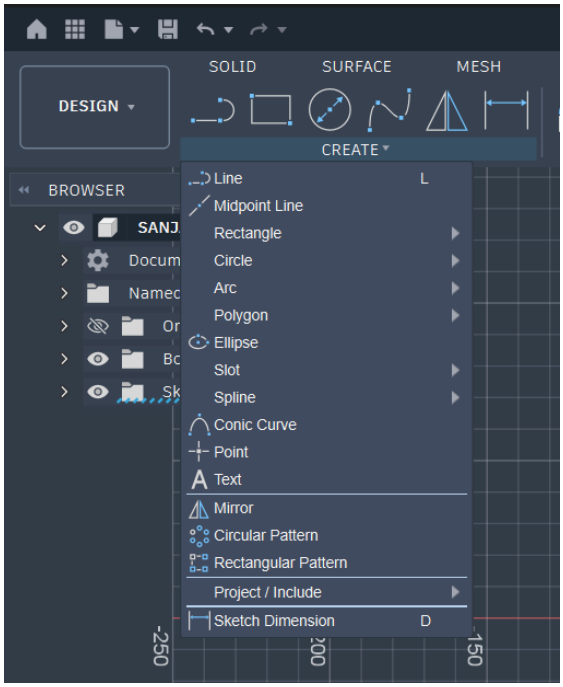
To sketch the given drawing with dimensions using fusion 360 tool.

REQUIREMENTS:

1. Autodesk fusion 360
2. Windows 11
3. 16 GB of RAM (integrated graphics recommend 2 GB or more)
4. 2.5 Mbps or faster download; 500 Kbps or faster upload

PROCEDURE:

1. Open a new file and then select the sketch icon.
2. Select the plane for sketching.
3. Use sketching features (line, circle, arc, rectangle, polygon, etc.,) to draw the given sketch.



Front View & Overall Lengths

- **Hole and Radius:** The main circular hole has a diameter of **16 millimeters** ($\emptyset 16$), with a matching radius of **16 millimeters** ($R16$) at the rounded end.

  Fork / Clevis Section: The length of the slotted fork body is **36 millimeters**.

- **Stem / Pin Section:** The cylindrical stem extending to the right measures **32 millimeters** in length, and its internal/through hole has a diameter of **18 millimeters** ($\emptyset 18$).

- **Total Length:** The complete overall length from the rounded end to the tip of the stem is **68 millimeters** ($36 + 32$).

Top View & Widths

- **Overall Length (Top):** The total length shown in the top view is **52 millimeters**, with a slot depth/recess of **11 millimeters** at the end.

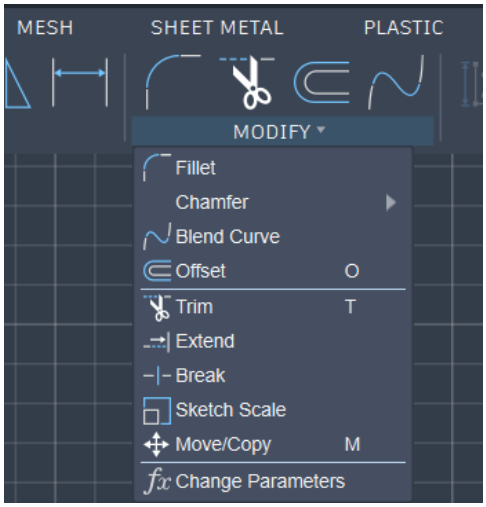
- **Width Spacing:** The thickness/width of the top and bottom fork flanges are both **11 millimeters**, leaving an inner gap of **18 millimeters** between them.

Right View & Cross-Section

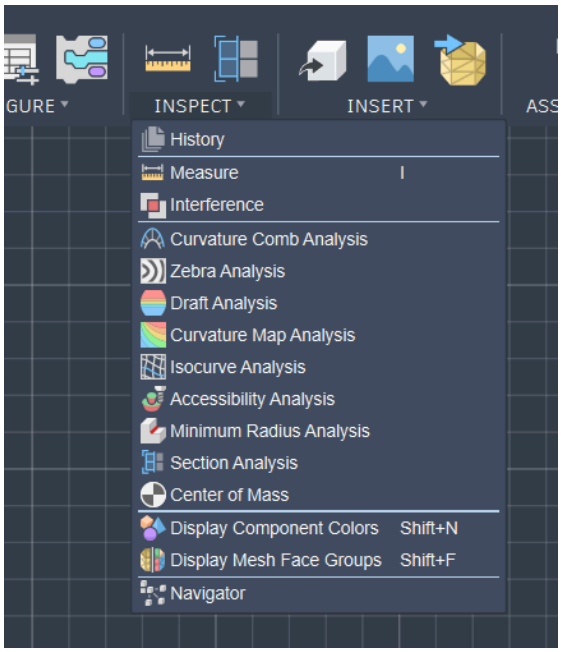
Square Flange / Base: The square block section measures **32 millimeters** by **32 millimeters** in height and width.

Depth / Thickness: The depth of this square flange section is **40 millimeters**.

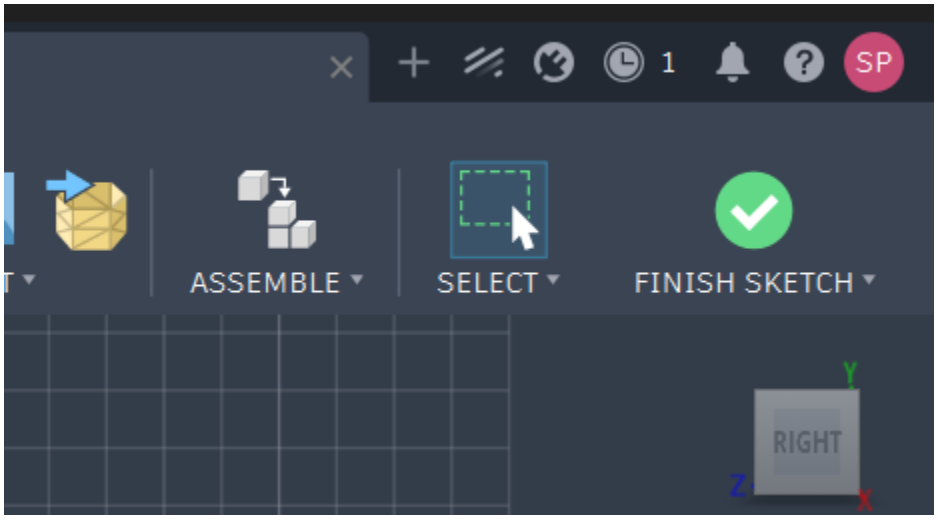
4. Use sketch modifying features (cut, trim, mirror, move, etc.,) to edit the sketch.



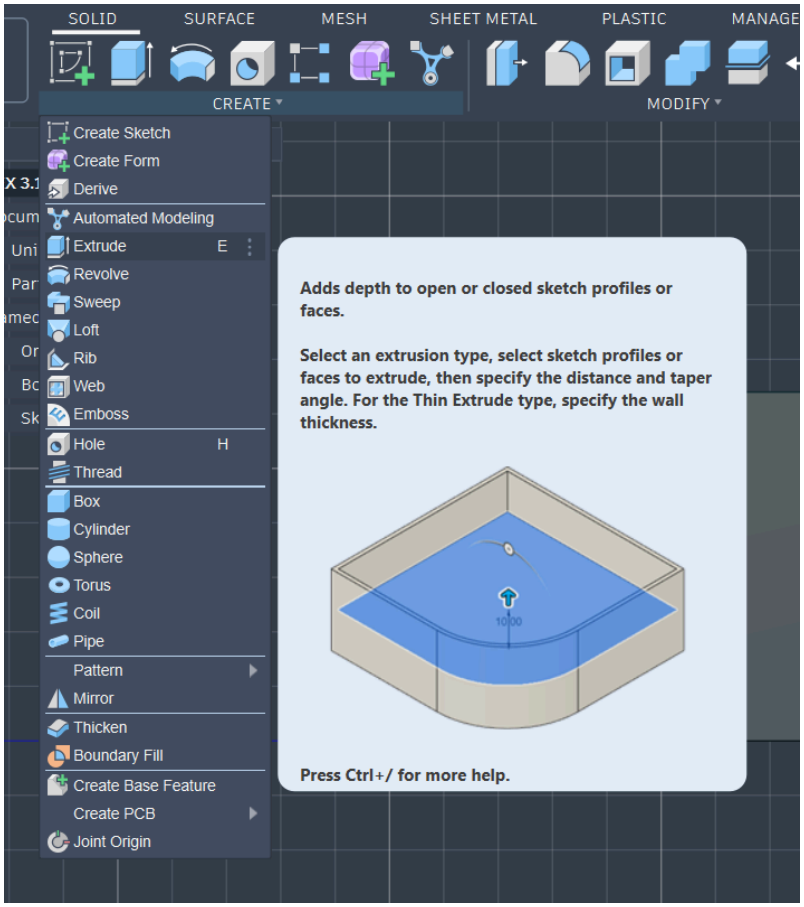
5. Use inspect feature to measure the dimensions of the sketch.



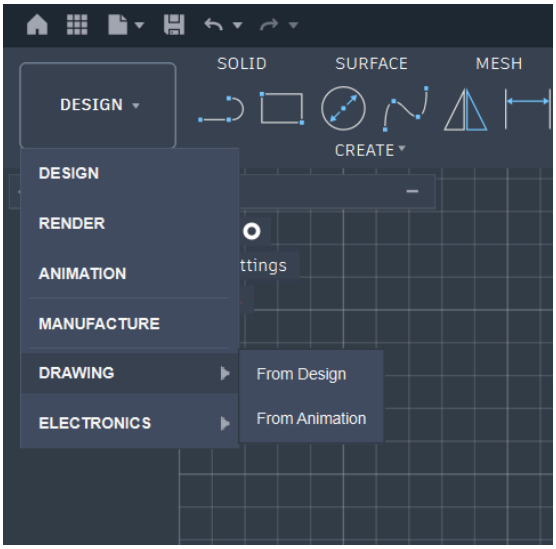
6. After finishing the sketch use the finish sketch option to finalize the sketch.



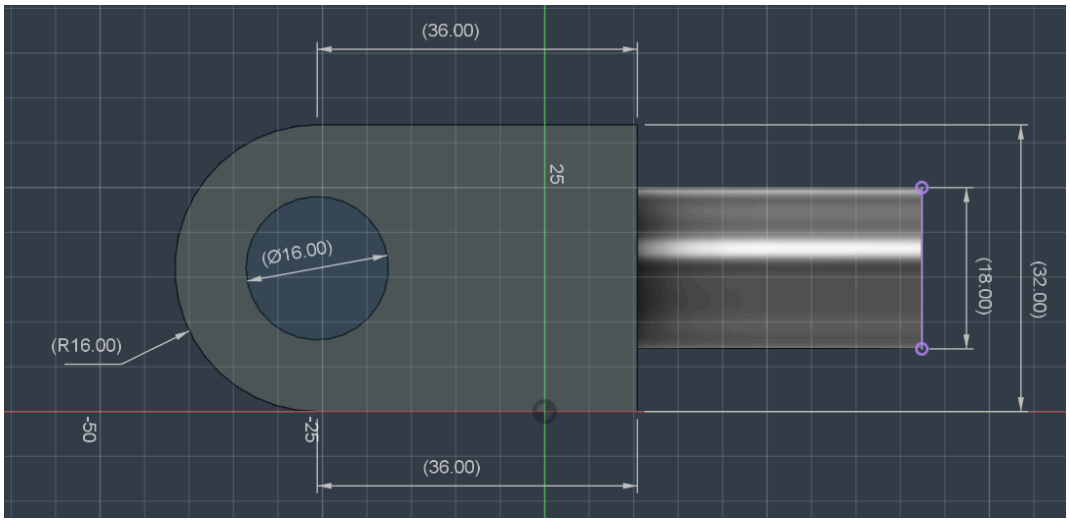
7. Use extrude option to convert 3D model from sketch

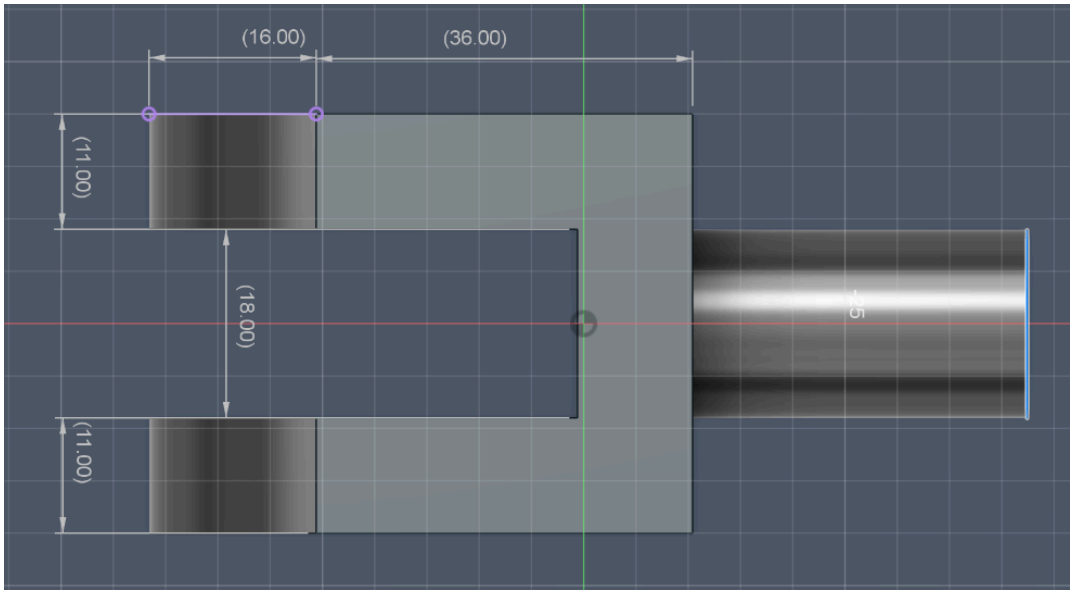


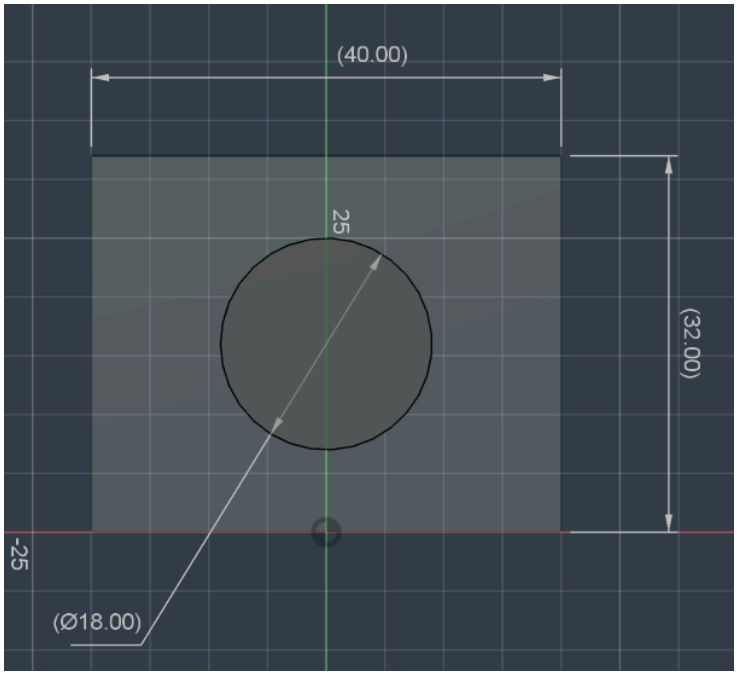
8. After finishing the model convert to drawing sheet



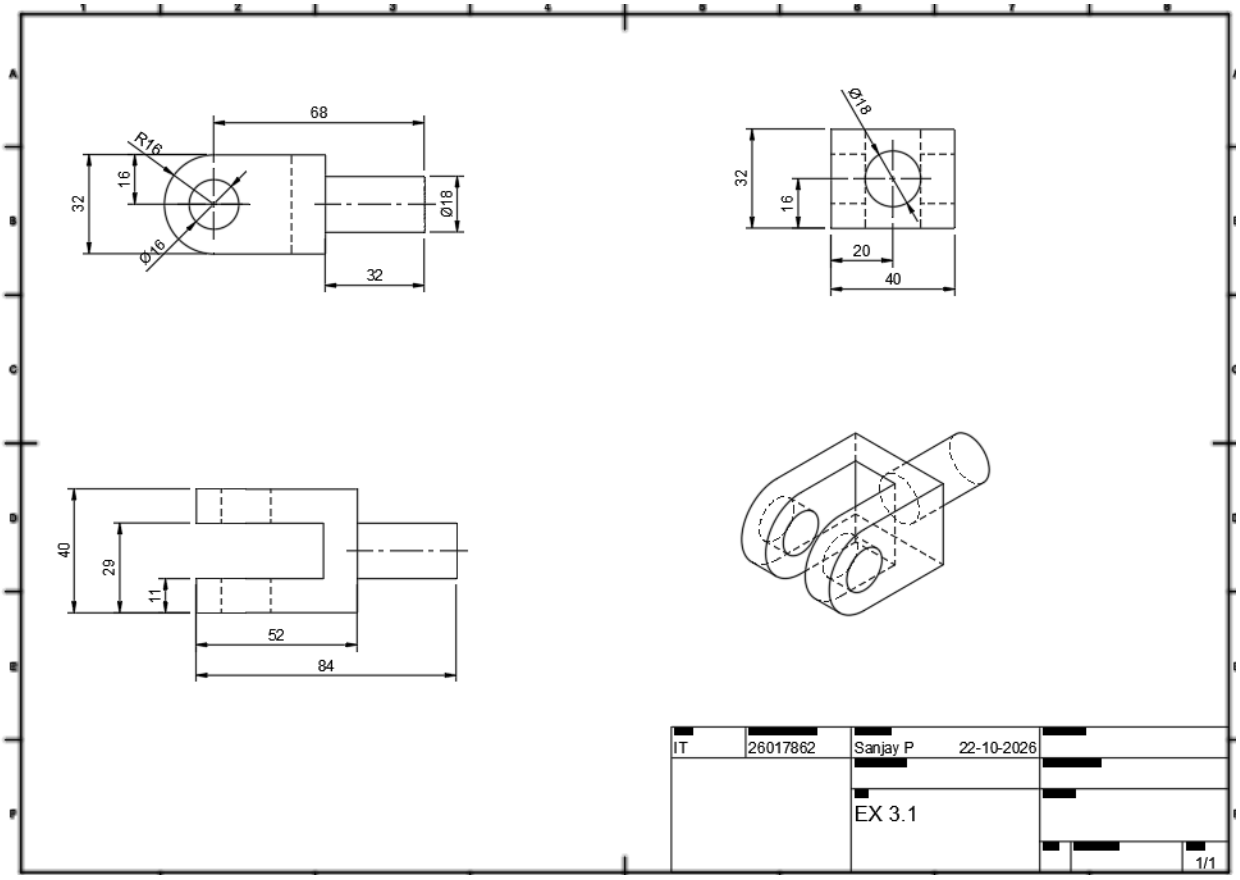
INPUT:




(16.00)
(36.00)
(11.00)
(18.00)
(11.00)
-15


(40.00)
25
(32.00)
-25
(Ø18.00)

OUTPUT:



RESULT:

Thus, the 3D model is created using orthographic projection.